Multivariable Calculus

Unit 3

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3 Multivariate Limits and Derivatives

3.1 Graphs of Multivariate Functions

- The **level curves** of a surface $z = f(x, y)$ are the curves of intersection at the level planes $z = k$, or in other words, the curves defined by $f(x, y) = k$. A level curve may not exist for all values of k.
- Similarly, the **level surfaces** of a volume $w = f(x, y, z)$ are the surfaces defined by $f(x, y, z) = k$.
- The **contour plot** is a way of representing a 3D surface on a 2D plane, by drawing all level curves on one plane. In other words, the contour plot of a surface $z = f(x, y)$ can be drawn by drawing lines which join the points (x, y) where $f(x, y)$ is equal.

3.2 Limits and Continuity in Multiple Variables

Summary

- **To show a limit does not exist,** check multiple paths and show result is $\neq$
- **To show and find a limit exists:**
  - Direct substitution (if function is continuous over its domain)
  - Algebraic manipulation (cancel factors/multiply by conjugates)
  - Squeeze Theorem
  - Convert to polar/spherical coordinates, then check if limit is constant
    $(x, y) \to (0, 0)$ becomes $r \to 0^+$, $\quad (x, y, z) \to (0, 0, 0)$ becomes $\rho \to 0^+$
  - Use $\delta - \varepsilon$ proof

Limits & Continuity

- Recall that for single-variable functions: $\lim_{x \to a} f(x) = L$ means that by approaching $x = a$ from both the left and right, $f(x)$ approaches the same value L. When $\lim_{x \to a} f(x) = f(a)$, the function is *continuous*.
- For a function of two variables $f(x, y)$, we can approach some point $(x, y) = (a, b)$ from an infinite number of paths. If f approaches the same limit L, then $\lim_{(x,y) \to (a,b)} f(x, y) = L$.
  - To show a limit exists, we must prove that by approaching the point from all paths, we get the same limit. If even one of the paths yield a different limit than another path, the limit does not exist.
  - f is **continuous** at (a, b) if $\lim_{(x,y) \to (a,b)} f(x, y) = f(a, b)$.

    We can use this definition in reverse to find limits of continuous functions (i.e. if we know a function is continuous and defined at (a, b), then we can just plug in (a, b) into the function to find its limit)

Example: Show that $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2+\sin^2 y}{2x^2+y^2}$ does not exist.

Check that approaching the limit from at least two paths yield different values:

From the line $y = 0$:

$$\lim_{\substack{(x,y)\to(0,0)\\ y\to 0}} \frac{x^2+\sin^2 y}{2x^2+y^2} = \lim_{x\to 0}\frac{x^2}{2x^2} = \frac{1}{2}.$$

From the line $x = 0$:

$$\lim_{\substack{(x,y)\to(0,0)\\ x\to 0}} \frac{x^2+\sin^2 y}{2x^2+y^2} = \lim_{y\to 0}\frac{\sin^2 y}{y^2} = \left(\lim_{y\to 0}\frac{\sin y}{y}\right)^2 = 1.$$

$$\therefore \lim_{(x,y)\to(0,0)} \frac{x^2+\sin^2 y}{2x^2+y^2} \text{ DNE.}$$

Squeeze Theorem

- Let there be three functions of two variables α, f, β.

  If, for a finite region D, $\alpha(x,y) \le f(x,y) \le \beta(x,y)$, then for any point (a,b) in D where $\lim\limits_{(x,y)\to(a,b)} \alpha(x,y) = \lim\limits_{(x,y)\to(a,b)} \beta(x,y) = L$, the **squeeze theorem** says that $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$.

Extension: Limits by Changing Coordinate Systems

- If our multivariate limit approaches the origin, we can convert our function into polar coordinates (for two variables) or spherical coordinates (for three variables).

  This decays our limit into a single-variable limit (the limit as distance from the origin approaches zero).

- For polar coordinates: as $(x,y)\to(0,0)$, $r\to 0^+$.
  For spherical coordinates: as $(x,y,z)\to(0,0,0)$, $\rho\to 0^+$.

Rectangular to Polar Coordinates
Let $g(r,\theta)$ be the polar transformation of $f(x,y)$, i.e. $g(r,\theta) = f(r\cos\theta, r\sin\theta)$. Then:

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{r\to 0^+} g(r,\theta).$$

Rectangular to Spherical Coordinates
Let $g(\rho,\theta,\varphi)$ be the spherical transformation of $f(x,y,z)$, i.e.
$g(\rho,\theta,\varphi) = f(\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$. Then:

$$\lim_{(x,y,z)\to(0,0,0)} f(x,y,z) = \lim_{\rho\to 0^+} g(\rho,\theta,\varphi).$$

- θ (or φ) is the angle of the path we approach the origin from. If our limit depends on θ or φ, our limit would depend on the angle of the path and thus our limit does not exist.

Example: Limit using Polar Coordinates

Determine if $\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ exists.

Converting into polar coordinates:

$$\frac{x^2 - y^2}{x^2 + y^2} \mapsto \frac{\cancel{r^2}(\cos^2\theta - \sin^2\theta)}{\cancel{r^2}} = \cos 2\theta.$$

As $(x, y) \to (0, 0)$, $r \to 0^+$. Thus, our limit is equivalent to:

$$\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{r\to 0^+} \cos 2\theta = \cos 2\theta.$$

The limit depends on θ, which is the angle of the path we approach the origin from. In other words, the limit is different depending on the path, meaning our limit does not exist.

Extension: Delta-Epsilon Definitions of Limits

The definition of a single-variable limit is:

$$\lim_{x\to a} f(x) = L \text{ means } \forall\varepsilon > 0, \exists\delta > 0 \text{ s.t. } 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

In other words, for any desired vertical distance ε, we must be able to find an x a horizontal distance δ away from $x = a$ where evaluating $f(x)$ is at most ε away from $y = L$.

We can extend this to 3D: if ε is the desired "error", then there must exist a point (x, y) a radial distance δ away from (a, b) where evaluating $f(x, y)$ is at most ε away from L.

Delta-Epsilon Definition of a Multivariate Limit

Let $f(x, y)$ be a function of two variables. If $\lim_{(x,y)\to(a,b)} f(x, y) = L$, then:

$$\forall\varepsilon > 0, \exists\delta > 0 \text{ s.t. } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

3.3 Partial Derivatives

- Recall that the derivative $f'(x)$ of a single-variable function $f(x)$ at some point $x = a$ is the rate of change of the output f as the input x changes near $x = a$.
- For a function of multiple independent variables, the variables can change independently. One variable can change at once, or multiple can change at the same time.
- The **partial derivative** $\frac{\partial f}{\partial x} = f_x(x, y, \dots)$ of a function of several variables $f(x, y, \dots)$ is the rate of change of f as only one of the variables (in this case x) changes, and the other variables are held constant.
  - If f is a function of x and y, the partial derivative w.r.t. x of f at a point $(x, y) = (a, b)$, $\frac{\partial f}{\partial x}\Big|_{(a,b)} = f_x(a, b)$ is the rate of change of f as x varies while other y is held constant near (a, b).
- **Notations:** $\frac{\partial f}{\partial x} = f_x = \partial_x f$ (Leibniz, Lagrange, and Euler notations respectively)

Limit Definition of a Partial Derivative

For Functions of Two Variables

The **partial derivative** of a function $f(x, y)$ with respect to x is:

$$\frac{\partial f}{\partial x} = f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

Similarly, the partial derivative of f with respect to y is:

$$\frac{\partial f}{\partial y} = f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}.$$

It is the change in f as the variable we are differentiating with respect to changes while all other variables are held constant.

General Case

Let $f(x_1, x_2, \dots, x_n)$ be a function of n variables. Then the **partial derivative** of f with respect to x_i is:

$$\frac{\partial f}{\partial x_i} = f_{x_i}(x_1, x_2, \dots, x_n) = \lim_{h \to 0} \frac{f(x_1, x_2, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}.$$

Example: Find the Partial Derivatives of $f(x, y, z) = xe^y + y^2 e^z + ze^{-x}$.

To find each partial derivative, differentiate normally, treating other variables constant.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(xe^y + y^2 e^z + ze^{-x}\right) = e^y + -ze^{-x}.$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(xe^y + y^2 e^z + ze^{-x}\right) = xe^y + 2ye^z.$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}\left(xe^y + y^2 e^z + ze^{-x}\right) = y^2 e^z + e^{-x}.$$

Higher-Order Partial Derivatives

A **second-order partial derivative** of a function of two variables $f(x, y)$ is the partial derivative of one of the partial derivatives of f. This means that there are **four** total second-order partial derivatives of f:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} \qquad f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}$$

Generally, a function of n variables has n^2 second-order partial derivatives.

Note

For Leibniz notation, the order of differentiation is read right to left, whereas with the subscript (Lagrange) notation, the order is from left to right.

Clairaut's Theorem

Suppose a function $f(x, y)$ is *defined* over a finite region D. If f_{xy} and f_{yx} are *continuous* throughout D, then for all points $(x, y) \in D$:

$$f_{xy} = f_{yx} \qquad \text{or} \qquad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}.$$

In other words, for continuous functions, the order of partial derivatives does not matter. This theorem can be extended to higher-order derivatives.

This property is often referred to as the **symmetry of second partial derivatives**.

- In a similar vein, a **third-order partial derivative** of a function is a partial derivative of one of the second-order partial derivatives of f. Generally, a function of n variables has n^3 third-order partial derivatives.
  e.g. $f(x, y)$ has 8 third-order partial derivatives: $f_{xxx}, f_{xxy}, f_{xyx}, f_{xyy}, f_{yxx}, f_{yxy}, f_{yyx}, f_{yyy}$.
- An nth-order partial derivative of a function of m variables is a partial derivative of one of the $(n-1)$th-order partial derivatives of f. Generally, a function of m variables has m^n nth-order partial derivatives.

Differentiability

Differentiability by Partial Derivatives

A function $f(x, y)$ is **differentiable** at a point (a, b) if f is both **continuous** on some closed finite region D which includes (a, b) and the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are both **defined** on the open region D.

This can be extended to more than 2 variables.

Partial Differential Equations

- An equation consisting of partial derivatives is called a **partial differential equation** (PDE). For example, $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ is a partial differential equation called the *wave equation*.
- A **solution** to a partial differential equation is a function which satisfies the equation.
  - To verify a solution, find necessary partial derivatives and plug them into the equation to see if it holds true.

Example: Verifying a solution to a PDE

Verify that the function $u(x, y) = e^x \sin y$ is a solution to the PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Find the necessary partial derivatives:

$$\frac{\partial u}{\partial x} = e^x \sin y \qquad \frac{\partial u}{\partial y} = e^x \cos y$$

$$\frac{\partial^2 u}{\partial x^2} = e^x \sin y \qquad \frac{\partial^2 u}{\partial y^2} = -e^x \sin y.$$

Plugging into the PDE:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin y - e^x \sin y \overset{\checkmark}{=} 0.$$

3.4 Tangent Planes & Approximations

Recall that the derivative of a single-variable function can be interpreted as the *slope of the tangent line* to the curve. A function of two variables defines a surface, there are an infinite number of tangent lines to the surface at any given point, however there exists a *plane* that consists of all tangent lines. This plane is called the **tangent plane**.

Consider a curve defined by single-variable function $y = f(x)$. Recall that the linearization of f at a point $x = x_0$ can be used to approximate f at points near x_0. Such linearization is:

$$y - y_0 = f'(x)(x - x_0).$$

Tangent Plane to a Surface

Let the surface S be defined by the equation $z = f(x, y)$. Then the **tangent plane** to S at a point $(x_0, y_0, f(x_0, y_0))$ is the plane defined by:

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

We can use this tangent plane to derive a general function called the *linearization function* which can be used to approximate values of f near the point:

Linearization of a Function of Two Variables

The **linearization** of a function $f(x, y)$ at (x_0, y_0) is:

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The **linear approximation** of a function $f(x, y)$ at a point P is the linearization at that point.

Example: Linear Approximation

Use the linearization of $f(x, y) = \sqrt{x^2 + y^2}$ at $(1, 1)$ to approximate $f(1.1, 1.2)$.

Find the partial derivatives of f:

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \qquad \frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

Plugging into each at $(1, 1)$ yields $f_x(1, 1) = f_y(1, 1) = \frac{\sqrt{2}}{2}$.

Find the linearization using the formula above:

$$f(x, y) \approx L(x, y) = \sqrt{2} + \frac{\sqrt{2}}{2}(x - 1) + \frac{\sqrt{2}}{2}(y - 1).$$

Thus:

$$f(1.1, 1.2) \approx L(1.1, 1.2) = \sqrt{2} + \frac{\sqrt{2}}{2}(0.1) + \frac{\sqrt{2}}{2}(0.2) = \frac{23\sqrt{2}}{20}.$$

Differentials

Recall the tangent plane equation. If the point of interest (x_0, y_0, z_0) is made infinitesimally close to an arbitrary point (x, y, z), then the difference of each component is also infinitesimally small. That is, if $x - x_0 \to 0$, we can call this difference $\mathrm{d}x$, called the **differential** of x.

Thus, our equation becomes:

$$\mathrm{d}z = f_x\,\mathrm{d}x + f_y\,\mathrm{d}y = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y.$$

So if $z = f(x, y)$, the above expression $\mathrm{d}z$ is the **total differential** of z.

Total Differential

Let f be a function of n variables: $f(x_1, x_2, \ldots, x_n)$. Then the **total differential** of f is:

$$\mathrm{d}f = \frac{\partial f}{\partial x_1}\,\mathrm{d}x_1 + \frac{\partial f}{\partial x_2}\,\mathrm{d}x_2 + \ldots + \frac{\partial f}{\partial x_n}\,\mathrm{d}x_n.$$

Example: Find the total differential of $V(r, h) = \frac{1}{3}\pi r^2 h$.

The partial derivatives of V are:

$$\frac{\partial V}{\partial r} = \frac{2}{3}\pi r h \qquad\qquad \frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2$$

Thus, the total differential is:

$$\mathrm{d}V = \frac{2}{3}\pi r h\,\mathrm{d}r + \frac{1}{3}\pi r^2\,\mathrm{d}h.$$

Using a finite interpretation of a differential, we interpret let each differential as an "error" or finite difference. Then, the total differential becomes an equation relating relatively small changes in each of the variables to the total change in function.

Example: Interpreting differentials as finite errors

A ruler is used to measure the height of a cone to an accuracy of 3 mm. A separate ruler is used to measure the radius of the cone to an accuracy of 4 mm.

The cone is measured this way to have a height of 14 mm and a radius of 6 mm, resulting in an estimated volume of 168π mm^3. What is the approximate max error of the estimate?

Using the total differential from the other example, we can estimate the error $\mathrm{d}V$ from the measurement. Since $\mathrm{d}r$ can be interpreted as the "error in r", we can set $\mathrm{d}r = 4$ mm. Similarly, $\mathrm{d}h = 3$ mm. Finally, $r = 6$ mm and $h = 14$ mm.

Plugging all four values into the expression for $\mathrm{d}V$:

$$\Delta V_{\max} \approx \mathrm{d}V = \frac{2}{3}\pi(6\text{ mm})(14\text{ mm})(4\text{ mm}) + \frac{1}{3}\pi(6\text{ mm})^2(3\text{ mm}) = 260\pi\text{ mm}^3.$$

Differentials vs. Increments

The **increment** of a function f, represented Δf is the *true* change in f over some change in its variables $\vec{r}$, represented $\Delta\vec{r}$. A **differential** uses a tangent line to approximate the increment ($\mathrm{d}f \approx \Delta f$), and the approximation gets better as $\mathrm{d}\vec{r} \to \vec{0}$.

Tangent Planes to Parametric Surfaces

- Recall that a parametric surface is defined by a vector function of two variables, for example $\vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$.
- The tangent plane to a parametric surface at a point P can be found using the cross product of the two tangent vectors $\frac{\partial\vec{r}}{\partial u}$ and $\frac{\partial\vec{r}}{\partial v}$.
  - Originating from P, all tangent vectors can be thought of as "lying on the tangent plane"
  - Thus normal vector $\vec{n}$ to the tangent plane is orthgonal to any two tangent vectors at P. We can choose the two most convenient ones, $\frac{\partial\vec{r}}{\partial u}$ and $\frac{\partial\vec{r}}{\partial v}$:

$$\vec{n} = \frac{\partial\vec{r}}{\partial u} \times \frac{\partial\vec{r}}{\partial v}.$$

  - If $\vec{n} = \langle a, b, c\rangle$, then the equation of the tangent plane at point $P(x_0, y_0, z_0)$ is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ by the standard normal-point form of a plane.

Example: Tangent Plane to a Parametric Surface

Find the equation of the plane tangent to the surface defined by $\vec{r}(u,v) = \langle u\cos v, u\sin v, v\rangle$ at the point $\left(0, 1, \frac{\pi}{2}\right)$.

First, find the values of u and v at the point:

$$u\cos v = 0 \qquad\qquad u\sin v = 1 \qquad\qquad v = \frac{\pi}{2}.$$

From the first two equations, we can see that $u = 1$ and $v = \frac{\pi}{2}$.

Next, find the tangent vectors:

$$\frac{\partial\vec{r}}{\partial u} = \vec{r}_u = \langle \cos v, \sin v, 0\rangle \qquad\qquad \frac{\partial\vec{r}}{\partial v} = \vec{r}_v = \langle -u\sin v, u\cos v, 1\rangle.$$

Plugging in $u = 1$ and $v = \frac{\pi}{2}$ yields:

$$\vec{r}_u\left(1, \frac{\pi}{2}\right) = \langle 0, 1, 0\rangle \qquad\qquad \vec{r}_v\left(1, \frac{\pi}{2}\right) = \langle -1, 0, 1\rangle.$$

The normal vector is:

$$\vec{n} = \vec{r}_u \times \vec{r}_v = \langle 0, 1, 0\rangle \times \langle -1, 0, 1\rangle = \langle 1, 0, 1\rangle.$$

Finally, plug the point and normal vector into the point-normal form of a plane:

$$1(x - 0) + 0(y - 1) + 1\left(z - \frac{\pi}{2}\right) = 0 \qquad \text{or} \qquad x + z = \frac{\pi}{2}.$$

Extension: Quadratic Approximations

Recall: the quadratic approximation formula for a single-variable function $f(x)$ from a known point $(a, f(a))$ is just the second-degree Taylor polynomial centered at $x = a$:

$$f(x) \approx Q(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2.$$

We can extend this to functions of two variables:

$$\begin{aligned} f(x,y) \approx Q(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ + \frac{f_{xx}(a,b)}{2}(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{f_{yy}(a,b)}{2}(y-b)^2. \end{aligned}$$

- The first term is just the function plugged in at the point, just like the single-variable approximation.
- The next two terms are just the linearization of f at (a, b). These are the parts of the approximation involving the "first-derivative", just like the single-variable approximation.
- The last three terms are the parts of the approximation involving the "second-derivative". There are actually four terms, but since $f_{xy} = f_{yx}$, the two mixed partial derivative terms can be combined (hence why there is no division by 2)

3.5 Chain Rule

The **chain rule** computes the derivative of a function of other functions.

- For single-variable functions $y = f(x)$, $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ if there is an "intermediate function" $u(x)$ where y is a function of u.
- For multivariate functions $f(x, y, \ldots)$, there are two "cases" of composite functions:
  - **Case 1:** if each variable is a function of a single variable, e.g. $x = g(t), y = h(t)$, then:
$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{df}{dy}\frac{dy}{dt} + \ldots$$
  - **Case 2:** if each variable is a function of several variables, e.g. $x = g(u, v), y = h(u, v)$, then f has partial derivatives in both u and v:
$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u} + \ldots$$
$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v} + \ldots$$

Example: Chain Rule

Let $z = x^2 + y^2 + xy$, where $x = \ln t$ and $y = \cos t$. What is $\frac{dz}{dt}$?

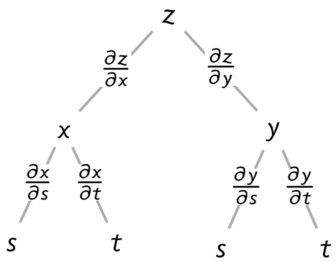
By the formula for case 1:
$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

The partial derivatives of z are:
$$\frac{\partial z}{\partial x} = 2x + y \qquad \frac{\partial z}{\partial y} = 2y + x.$$

The derivatives of x and y are:
$$\frac{dx}{dt} = \frac{1}{t} \qquad \frac{dy}{dt} = -\sin t.$$

Thus:
$$\frac{dz}{dt} = (2x + y)\left(\frac{1}{t}\right) + (2y + x)(-\sin t) = \frac{2\ln t + \cos t}{t} - (2\cos t + \ln t)\sin t.$$

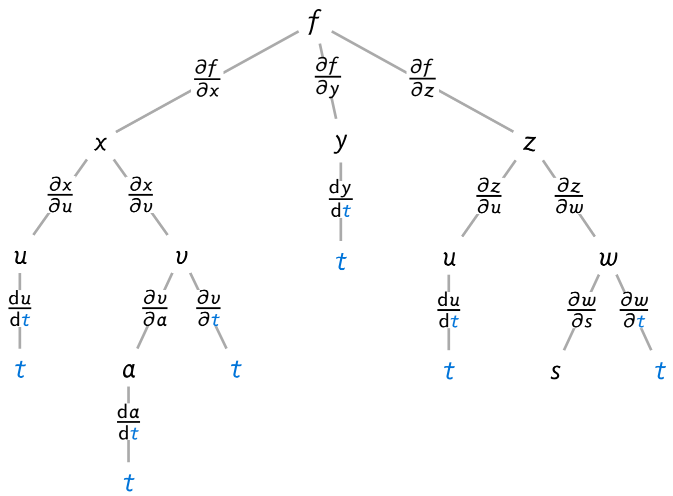
- We can use a **tree diagram** to help visualize the chain rule. Each node is a variable, and each edge represents a partial derivative.
  - The path from the root to a leaf node represents the full "chain" between variables.
  - Find all paths that connect the root to the leaf node of interest, and sum the products of each path.



Example: Chain Rule for Tree Diagram

Use a tree diagram to find a formula for $\frac{\partial f}{\partial t}$ if the following functions are given: $f(x, y, z)$, $x(u, v)$, $y(t)$, $z(u, w)$, $u(t)$, $v(\alpha, t)$, $\alpha(t)$, and $w(s, t)$.

Find all nodes with t, then trace each path leading to each node:



There are 6 paths from f to t:

1. $f \to x \to u \to t$ becomes the "chain" $\frac{\partial f}{\partial x}\frac{\partial x}{\partial u}\frac{du}{dt}$.
   "f is a function of x, which is a function of u, which is a function of t"
2. $f \to x \to v \to \alpha \to t$
3. $f \to x \to v \to t$
4. $f \to y \to t$
5. $f \to z \to u \to t$
6. $f \to z \to w \to t$

Thus, by the chain rule:

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u}\frac{du}{dt} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial v}\frac{\partial v}{\partial \alpha}\frac{d\alpha}{dt} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial v}\frac{\partial v}{\partial t} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial u}\frac{du}{dt} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial w}\frac{\partial w}{\partial t}.$$

Implicit Differentiation

- A function can be either **explicitly** defined (e.g. $z = f(x, y)$) or **implicitly** defined (e.g. $F(x, y, z) = 0$, where z is a function of x and y).
  - For example, $\sin(y) = x^2$ implicitly defines y as a function of x.
- To differentiate an implicitly defined function, we use the chain rule just like with single-variable functions.
  - Recall: if $y(x)$ is implicitly defined by $\sin(y) = x^2$, we can find $\frac{dy}{dx}$ by:
    1. taking $\frac{d}{dx}$ of both sides: $\frac{d}{dx}\sin(y) = 2x$
    2. using the chain rule if we see a y: $\frac{d}{dx}\sin(y) = \frac{d(\sin(y))}{dy}\frac{dy}{dx} = \cos(y)\frac{dy}{dx}$
    3. solving for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{2x}{\cos(y)}$.
  - If $z(x, y)$ is implicitly defined, then there are two derivatives we can find: $\frac{\partial z}{\partial x}$ or $\frac{\partial z}{\partial y}$.

- To find $\frac{\partial z}{\partial x}$, take $\frac{\partial}{\partial x}$ of both sides, use the chain rule when you see a z, then solve for $\frac{\partial z}{\partial x}$.
- To find $\frac{\partial z}{\partial y}$, take $\frac{\partial}{\partial y}$ of both sides, use the chain rule when you see a z, then solve for $\frac{\partial z}{\partial y}$.

Example: Implicit Differentiation

Let $z(x, y)$ be a function of x and y. If $x^3 + y^3 + z^3 + 6xyz = 1$, what is $\frac{\partial z}{\partial x}$?

To find $\frac{\partial z}{\partial x}$, take the partial derivative w.r.t. x of both sides:

$$\frac{\partial}{\partial x}(x^3 + y^3 + z^3 + 6xyz) = \frac{\partial}{\partial x}(1)$$
$$3x^2 + 0 + \frac{\partial}{\partial x}(z^3) + \frac{\partial}{\partial x}(6xyz) = 0.$$

Let $u = z^3$. To find $\frac{\partial}{\partial x}(z^3) = \frac{\partial u}{\partial x}$, we must use the chain rule, since u is not an independent variable. Since u is a function of z is a function of x, the chain rule tells us that:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial z}\frac{\partial z}{\partial x}.$$

Since $\frac{\partial u}{\partial z} = 3z^2$, $\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(z^3) = 3z^2\frac{\partial z}{\partial x}$. Similarly we can find using the product rule that $\frac{\partial}{\partial x}(6xyz) = 6y\frac{\partial}{\partial x}(xz) = 6yz + 6xy\frac{\partial z}{\partial x}$:

$$3x^2 + 3z^2\frac{\partial z}{\partial x} + 6yz + 6xy\frac{\partial z}{\partial x} = 0$$
$$\frac{\partial z}{\partial x} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}.$$

Implicit Function Theorem

We can also use partial differentiation directly to find the derivative of implicitly-defined functions (alternative to implicit differentiation).

Let $y(x)$ be implicitly defined by the equation $F(x, y) = 0$. By taking the partial derivative of both sides w.r.t. x, we get:

$$\frac{\partial}{\partial x}(F(x, y)) = 0$$
$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\partial F/\partial x}{\partial F/\partial y}.$$

Implicit Function Theorem for Single-Variable Functions

Let $y = f(x)$ be defined implicitly by the equation $F(x, y) = 0$. Then:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\partial F/\partial x}{\partial F/\partial y}.$$

Example: If $x^2 + y^2 + xy = 7$, what is $\frac{dy}{dx}$?

Let $F(x, y) = x^2 + y^2 + xy - 7$. The partial derivatives of F are:

$$\frac{\partial F}{\partial x} = 2x + y \qquad \frac{\partial F}{\partial y} = 2y + x.$$

Thus:

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} = -\frac{2x + y}{2y + x}.$$

If we have a function $z(x, y)$ implicitly defined by the equation $F(x, y, z) = 0$, we can find both $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ by taking the partial derivative of both sides w.r.t. x and y, respectively:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0 \implies \frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z}.$$

$$\frac{\partial F}{\partial x}\frac{\partial x}{\partial y} + \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial y} = 0 \implies \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}.$$

Since x is independent of y, $\frac{\partial y}{\partial x} = 0$ and $\frac{\partial x}{\partial y} = 0$.

Implicit Function Theorem for Multivariable Functions

Let $z = f(x, y)$ be defined implicitly by the equation $F(x, y, z) = 0$. Then:

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} \qquad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}.$$

Example: Implicit Function Theorem for Multivariable Functions

Let $z(x, y)$ be defined implicitly by the equation $x^3 + y^3 + z^3 + 6xyz = 1$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Let $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1$. The partial derivatives of F are:

$$\frac{\partial F}{\partial x} = 3x^2 + 6yz \qquad \frac{\partial F}{\partial y} = 3y^2 + 6xz \qquad \frac{\partial F}{\partial z} = 3z^2 + 6xy.$$

Thus:

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} \qquad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z} = -\frac{3y^2 + 6xz}{3z^2 + 6xy}.$$

This is the same answer we got using implicit differentiation.

3.6 Gradients

The **gradient** is a vector of all partial derivatives of a function. If f is a function that takes in n parameters, then the gradient of f, typically denoted $\vec{\nabla} f$, is an n-dimensional vector.

Gradient of a Function

Let f be a function of n variables: $f(x_1, x_2, \dots, x_n)$. Then the **gradient** of f is:

$$\vec{\nabla} f(x_1, x_2, \dots, x_n) = \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

For example, $\vec{\nabla} f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$ and $\vec{\nabla} f(x, y, z) = \langle f_x, f_y, f_z \rangle$.

The *nabla* operator, $\vec{\nabla}$, can be interpreted as a vector of partial derivative operators:

$$\vec{\nabla} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This is why $\vec{\nabla} f$ can be used to represent the gradient of f:

$$\vec{\nabla} f = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

This definition of $\vec{\nabla}$ will be helpful in future units.

Total Differential in terms of the Gradient

Take the total differential of a function $f(x, y)$, $\mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y$. We can represent this as the following dot product:

$$\mathrm{d}f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \cdot \langle \mathrm{d}x, \mathrm{d}y \rangle.$$

If we let $\mathrm{d}\vec{r} = \langle \mathrm{d}x, \mathrm{d}y \rangle$ (a change in "position"), then the total differential can be written as just:

$$\mathrm{d}f = \vec{\nabla} f \cdot \mathrm{d}\vec{r}.$$

Directional Derivatives

- Suppose a surface $z = f(x, y)$ is graphed in the 3D-plane. Each partial derivative is the slope of the tangent line in the direction of the respective axis. For example, $\frac{\partial z}{\partial x}$ is the slope of the tangent line in the direction of $+x$.
- With partial derivatives, we can only represent the slope in two directions ($+x$ and $+y$). To represent the slope in any direction, we use a **directional derivative**
- The directional derivative $D_{\vec{v}}\, f(x, y)$ is the slope of the line tangent to the surface *in the direction* of $\vec{v}$ is denoted $D_{\vec{v}} f(x, y)$.

Directional Derivative

Let $z = f(x,y)$ be a surface in the 3D-plane, and $\hat{v} = \langle v_1, v_2 \rangle$ is a 2D unit vector. The **directional derivative** in the direction of $\hat{v}$ is:

$$D_{\hat{v}}\, f(x,y) = \vec{\nabla} f(x,y) \cdot \hat{v} = \frac{\partial f}{\partial x} v_1 + \frac{\partial f}{\partial y} v_2.$$

- Notice that the definition above only works when $\vec{v}$ is a unit vector.
- An n-dimensional manifold has a directional derivative in the direction of an $(n-1)$-dimensional vector (e.g. a 3D shape has a 2D dimensional gradient)
- The notation $\nabla_{\vec{v}}$ is sometimes used to denote $D_{\vec{v}}$. (e.g. $\nabla_{\vec{v}} f$).

Properties of Gradients

Direction of Steepest Ascent

- The direction for which the directional derivative is highest (direction of **steepest ascent**) is in the direction of the gradient vector. The maximum value of the directional derivative is the magnitude of the gradient:

$$\max D_{\hat{v}}\, f(x,y) = \left|\vec{\nabla} f(x,y)\right|.$$

- In the same way, the direction of **steepest descent** is in the direction of $-\vec{\nabla} f(x,y)$, and the minimum value of the directional derivative is $-\left|\vec{\nabla} f(x,y)\right|$.
- If the angle between the direction vector and the gradient vector is θ, then by the definition of the dot product,

$$D_{\hat{v}}\, f(x,y) = \left|\vec{\nabla} f(x,y)\right| |\hat{v}| \cos\theta = \left|\vec{\nabla} f(x,y)\right| \cos\theta.$$

  - Note: this θ is **not** the angle between the x-axis and the direction vector!

Normal Lines and Tangent Planes

- The gradient vector at a point is **orthogonal** (perpendicular) to the level curve of the function at that point (or level surface if f is a function of 3 variables).
  - A level curve of f is in the form $f(x,y) = c$, where c is a constant. A level surface of f is in the form $f(x,y,z) = c$.
- The **tangent line** to a level *curve* using $\vec{\nabla} f$ is $\vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) = 0$.
- We can find the **normal line** and **tangent plane** to a level *surface* using $\vec{\nabla} f$:
  - The normal line at point $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ is given by:

$$\vec{r} = \vec{r}_0 + t\, \vec{\nabla} f(\vec{r}_0).$$

  - The tangent plane at $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ is given by:

$$\vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) = 0.$$

Generalization of a Linear Approximation

Using the gradient vector, we can generalize the linear approximation for a multivariate function f, given its "center" $\vec{r}_0 = \langle x_0, y_0, z_0, \ldots \rangle$ (i.e. the "given point"):

$$L(\vec{r}) = f(\vec{r}_0) + \vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0).$$

This is a first-order Taylor polynomial for f. It is similar to the formula for single-variable functions, but instead of derivatives, gradients are used.

Note that we pass in a position vector $\vec{r} = \langle x, y, z, \ldots \rangle$ to represent all parameters passed into f.

Extension: The Hessian Matrix

If represent $\vec{\nabla}$ as a column vector and perform matrix multiplication:

Hessian Operator (Gradient Definition)

The **hessian operator H**, when applied to a function, returns a matrix of its second-order partial derivatives. The operator is defined as:

$$\mathbf{H} = \vec{\nabla}\ \vec{\nabla}^{\mathsf{T}}.$$

For example, the Hessian operator applied to a function $f(x, y)$ is:

$$\mathbf{H}(f) = \vec{\nabla}\ \vec{\nabla}^{\mathsf{T}} f = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \begin{pmatrix} \partial_x & \partial_y \end{pmatrix} f = \begin{pmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{pmatrix}.$$

The resulting matrix is known as the **Hessian matrix** of f:

Hessian Matrix

Let $f(x_1, x_2, \ldots, x_n)$ be a function of n variables. Then the **Hessian matrix** of f is the matrix of all partial derivatives:

$$\mathbf{H}(f) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

Or, more compactly, it is the $n \times n$ matrix $\mathbf{H}$ where $\mathbf{H}_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

The Hessian matrix can be thought of as a "second-order gradient" and the analog to the second derivative in the Taylor polynomial.

Thus, the quadratic approximation of f at $\vec{r}_0$ is:

$$Q(\vec{r}) = f(\vec{r}_0) + \vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) + \frac{1}{2} (\vec{r} - \vec{r}_0)^{\mathsf{T}} \mathbf{H}(f(\vec{r}_0)) (\vec{r} - \vec{r}_0).$$

3.7 Extreme Values

Local Extrema

- A function $f(x, y)$ has a *local maximum* at (a, b) if $f(a, b) \geq f(x, y)$ for all points (x, y) near (a, b).
- A function $f(x, y)$ has a *local minimum* at (a, b) if $f(a, b) \leq f(x, y)$ for all points (x, y) near (a, b).
- Collectively, these are called *local extrema*.

Just like univariate functions, multivariate functions can only attain local extrema at **critical points**. This is where the function has a derivative that is either zero or undefined:

> **Critical Points**
>
> Let f be a function of several variables: $f(x, y, \ldots)$. f has a **critical point** at $(a, b, \ldots)$ if $\vec{\nabla} f(a, b, \ldots) = \vec{0}$ or it is not differentiable at $(a, b, \ldots)$ (i.e. $\vec{\nabla} f$ is undefined).

A local extrema does not exist at every critical point. To determine if a critical point is a local extrema, we can use the *second partial derivative test*:

> **Second Partial Derivative Test**
>
> Let $f(x, y)$ be a function of two variables whose second-order partial derivatives are **continuous**. Let $D = \det(\mathbf{H}(f))$ be the determinant of the *Hessian matrix* of f.
> In other words: $D = \begin{vmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$.
>
> If (a, b) is a **critical point** of f, then:
> - if $D(a, b) > 0$:
>   - ...and $f_{xx} > 0$, then there is a *local minimum* at (a, b).
>   - ...and $f_{xx} < 0$, then there is a *local maximum* at (a, b).
> - if $D(a, b) < 0$, there is a *saddle point* at (a, b).
> - if $D(a, b) = 0$, the test is *inconclusive*.

Example: Finding Local Extrema using Second Partial Derivative Test

Find and classify all local extrema of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 5$.

First, find the critical points by setting $\vec{\nabla} f = \vec{0}$:

$$\vec{\nabla} f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 3x^2 - 3, 3y^2 - 12 \rangle.$$

Setting each component equal to zero gives:

$$\begin{aligned} 3x^2 - 3 &= 0 & \qquad 3y^2 - 12 &= 0 \\ x &= \pm 1 & y &= \pm 2. \end{aligned}$$

Thus, the critical points are $(1, 2), (1, -2), (-1, 2), (-1, -2)$.

Next, compute the second-order partial derivatives:

$$f_{xx} = 6x \qquad f_{yy} = 6y \qquad f_{xy} = 0.$$

Thus, $D = |\mathbf{H}(f)| = \begin{vmatrix} 6x & 0 \\ 0 & 6y \end{vmatrix} = 36xy$.

Now evaluate D at each critical point:

(x, y)	$D(x, y)$	$f_{xx}(x, y)$	**Classification**
$(1, 2)$	36	6	Local Minimum since $D > 0$ and $f_{xx} > 0$
$(1, -2)$	-36		Saddle Point since $D < 0$
$(-1, 2)$	-36		Saddle Point since $D < 0$
$(-1, -2)$	36	-6	Local Maximum since $D > 0$ and $f_{xx} < 0$

Absolute Extrema

- A function $f(x, y)$ as an *absolute maximum* at (a, b) on a region D if $f(a, b) \geq f(x, y)$ for all points (x, y) in D.
- A function $f(x, y)$ as an *absolute minimum* at (a, b) on a region D if $f(a, b) \leq f(x, y)$ for all points (x, y) in D.
- Collectively, these are called *absolute extrema*.

Extreme Value Theorem

Let $f(x, y)$ be a function of two variables that is **continuous** on a *closed* and *bounded* region D. Then f attains both an absolute maximum and an absolute minimum on D.

Candidates Test

Recall the candidates test for single-variable functions: to find absolute extrema on a closed interval $[a, b]$:

1. Evaluate the function at all critical points in (a, b)
2. Evaluate the function at the endpoints $x = a$ and $x = b$.
3. The largest value is the absolute maximum, and the smallest value is the absolute minimum.

For multivariate functions $f(x, y)$ on a closed and bounded region D:

1. Evaluate the function at all critical points in the interior of D (where $\vec{\nabla} f = \vec{0}$ or undefined).
2. Evaluate the function on all candidates on the *boundary* of D.
3. The largest value is the absolute maximum, and the smallest value is the absolute minimum.

Essentially, the absolute extrema can occur either in the interior of the region (which we can easily find candidates for using $\vec{\nabla} f = \vec{0}$) or on the boundary of the region. Unlike single-variable functions, there are an infinite number of points at the boundary.

We must find a way to find candidates on the boundary which *may* contain the extrema.

For step two, there are two primary methods of finding these candidates:

- Find the absolute extrema of f on parts of the boundary where we can reduce f to a single-variable function (e.g. along line segments) — this is called the *closed interval method*.
- Use the method of **Lagrange Multipliers** to find candidates on more complex boundaries (this is the next section)

Example: Closed Interval Method with Line Segments

Find the absolute extrema of $f(x, y) = x^2 - 2xy + 2y$ on the triangular region with vertices at $(0, 0), (2, 0), (0, 2)$.

First, find the critical points in the interior of the region by setting $\vec{\nabla} f = \vec{0}$:

$$\vec{\nabla} f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 2x - 2y, -2x + 2 \rangle.$$

This is a linear system with only one solution: $(1, 1)$. This is our first *candidate*.

Next, evaluate f on each side of the triangle (L_1, L_2, and L_3):

- L_1 is the line segment from $(0, 0)$ to $(2, 0)$. Here, $y = 0$, so $f(x, 0) = x^2$.

  To find the extrema on L_1 we consider critical points *and* endpoints as candidates.
  - $f(x, 0)$ has a critical point at $x = 0$ since it satisfies $\frac{d}{dx} f(x, 0) = 2x = 0$. We must also consider endpoints at $x = 0$ and $x = 2$. Add $(0, 0), (2, 0)$ as candidates.

- L_2 is the line segment from $(2, 0)$ to $(0, 2)$. Here, $y = -x + 2$, so $f(x, -x + 2) = x^2 - 2x(-x + 2) + 2(-x + 2) = 3x^2 - 6x + 4$.

  There is a critical point at $x = 1$ ($\frac{d}{dx} f(x, -x + 2) = 6x - 6 = 0 \implies x = 1$) and endpoints at $x = 0$ and $x = 2$. Add $(1, 1), (2, 0), (0, 2)$ as candidates.

- L_3 is the line segment from $(0, 2)$ to $(0, 0)$. Here, $x = 0$, so $f(0, y) = 2y$.

  There is no critical point and endpoints at $y = 0$ and $y = 2$. Add $(0, 0), (0, 2)$ as candidates.

Now evaluate f at all candidates:

Source	(x, y)	$f(x, y)$
$\vec{\nabla} f = \vec{0}$ and L_2	$(1, 1)$	1
L_1, L_3	$(0, 0)$	0
L_1, L_2	$(2, 0)$	4
L_2, L_3	$(0, 2)$	4

The absolute minimum is 0 at $(0, 0)$ and the absolute maximum is 4 at $(2, 0)$ and $(0, 2)$.

3.8 Lagrange Multipliers

- When trying to find absolute extrema in a closed region, we had to find the minima and maxima of a known *boundary*.
  - For boundaries of single variable functions, all we need is a left boundary and a right boundary, so we only have to test two values
  - For boundaries of multivariate functions, there could be an infinite number of points on the boundary.
- The *closed interval method* works when the boundary can be reduced to single-variable functions (e.g. line segments), allowing us to determine a finite set of *candidates* for which we can test the function at to find the absolute extrema.
  - However, this method does not work for more complex boundaries (e.g. circles, ellipses, curves, etc.).
- The method of **Lagrange Multipliers** allows us to achieve this step. It is an alternative method of finding a finite set of *candidates* on the boundary.

Method of Lagrange Multipliers with One Constraint

Intuition

- Suppose we want to find the absolute extrema of a function $f(x, y)$ subject to a constraint $g(x, y) = c$ (where c is a constant). The constraint is a level curve of g.
- At the points where f attains its absolute extrema on the constraint, the level curve of f is *tangent* to the level curve of g.
  - This means that the two curves have the same slope at these points.
- The gradients of f and g are orthogonal to their respective level curves. Thus, at the points where f attains its absolute extrema on the constraint, the gradients of f and g are *parallel*.
  - In other words, there exists a scalar λ (called a **Lagrange multiplier**) such that:

$$\vec{\nabla} f(x, y) = \lambda \, \vec{\nabla} g(x, y).$$

Procedure

1. The first constraint is that the gradients of f and g are parallel: $\vec{\nabla} f(x, y) = \lambda \vec{\nabla} g(x, y)$
   This can be rewritten as two equations:

$$f_x = \lambda g_x \qquad\qquad f_y = \lambda g_y.$$

2. The second constraint is the original constraint: $g(x, y) = c$.

   Together, we have a system of three scalar equations with three unknowns (x, y, λ):

$$\begin{aligned} f_x &= \lambda g_x \\ f_y &= \lambda g_y \\ g(x, y) &= c. \end{aligned}$$

3. Solving this system gives us a finite number of *candidates* (x, y) for which we can evaluate $f(x, y)$ to find the absolute extrema subject to the constraint.

Method of Lagrange Multipliers with One Constraint

The *candidates* for the absolute extrema of a function $f(x, y)$ subject to the constraint $g(x, y) = c$ occur for all (x, y, λ) that satisfy the system of equations:

$$\vec{\nabla} f = \lambda \, \vec{\nabla} g$$
$$g(x, y) = c.$$

For a solution $(x, y, \ldots, \lambda)$, λ is called the **Lagrange multiplier** for the candidate $(x, y, \ldots)$.

Example: Lagrange Multipliers with One Constraint

Find the absolute extrema of $f(x, y) = x^2 + y^2$ subject to the constraint $x + y = 1$.

First, compute the gradients of f and $g(x, y) = x + y$:

$$\vec{\nabla} f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 2x, 2y \rangle \qquad \vec{\nabla} g(x, y) = \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right\rangle = \langle 1, 1 \rangle.$$

Next, set up the system of equations:

$$2x = \lambda \qquad 2y = \lambda \qquad x + y = 1.$$

Solving this system gives two candidates: $(0, 1)$ and $(1, 0)$.

Finally, evaluate f at each candidate:

(x, y)	$f(x, y)$
$(0, 1)$	1
$(1, 0)$	1

Thus, the absolute minimum is 1 at both $(0, 1)$ and $(1, 0)$. There is no absolute maximum since f is unbounded on the constraint.

We can combine this with the standard candidates test from the previous section to find absolute extrema on closed and bounded regions with more complex boundaries:

1. Find the critical points in the interior of the region by setting $\vec{\nabla} f = \vec{0}$ or where $\vec{\nabla} f$ is undefined.
2. Suppose the boundary of f is the constraint $g(x, y) = c$. Then, we can use the method of Lagrange multipliers to find candidates on the boundary.
3. Evaluate f at all candidates from the interior and boundary, and determine the absolute extrema.

Method of Lagrange Multipliers with Multiple Constraints

The method of Lagrange multipliers can be extended to work with multiple constraints.

1. Suppose we want to find the absolute extrema of a function $f(x, y, z)$ subject to two constraints: $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$ (where c_1 and c_2 are constants).

- Each constraint reduces the dimensionality of the domain by one. Thus, with two constraints, we are restricting f to a curve in 3D space. (Whereas with only one, we were restricting f to a surface in 3D space.)
- The constraint can also be thought of as the *curve of intersection* between the two surfaces defined by $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$.

2. At the points where f attains its absolute extrema on the constraints, the level surface of f is tangent to both constraint surfaces.
   - This means that the gradients of f, g, and h are all parallel.
     In other words, there exists scalars λ and μ such that:

$$\vec{\nabla} f(x, y, z) = \lambda \vec{\nabla} g(x, y, z) + \mu \vec{\nabla} h(x, y, z).$$

3. The second and third constraints are the original constraints: $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$. Together, we have a system of five scalar equations with five unknowns (x, y, z, λ, μ):

$$\begin{aligned} f_x &= \lambda g_x + \mu h_x \\ f_y &= \lambda g_y + \mu h_y \\ f_z &= \lambda g_z + \mu h_z \\ g(x, y, z) &= c_1 \\ h(x, y, z) &= c_2. \end{aligned}$$

Method of Lagrange Multipliers with Multiple Constraints

The *candidates* for the absolute extrema of a function $f(x, y, z)$ subject to the constraints $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$ occur for all (x, y, z, λ, μ) that satisfy the system of equations:

$$\begin{aligned} \vec{\nabla} f &= \lambda \, \vec{\nabla} g + \mu \, \vec{\nabla} h \\ g(x, y, z) &= c_1 \\ h(x, y, z) &= c_2. \end{aligned}$$

For a solution (x, y, z, λ, μ), λ and μ are called the **Lagrange multipliers** for (x, y, z).

Example: Absolute Extrema of a Function of Three Variables

Let $f(x, y, z) = x^2 + 2y^2 - 3xy + z^2$. The closed curve C is defined by the intersection of the surfaces defined by $x^2 + y^2 + z^2 = 6$ and $x + y + z = 0$. The region D is the set of all points on and inside the curve C. Find the absolute extrema of f on the region D.

First, find the critical points in the interior of the region by setting $\vec{\nabla} f = \vec{0}$.

$$\vec{\nabla} f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle = \langle 2x - 3y, 4y - 3x, 2z \rangle.$$

Setting each component equal to zero gives:

$$2x - 3y = 0 \qquad 4y - 3x = 0 \qquad 2z = 0$$
$$(x, y, z) = (0, 0, 0).$$

Thus, the only critical point in the interior is $(0, 0, 0)$. Next, use the method of Lagrange multipliers to find candidates on the boundary defined by the constraints $g(x, y, z) = x^2 + y^2 + z^2 = 6$ and $h(x, y, z) = x + y + z = 0$. The gradients of f, g, and h are:

$$\vec{\nabla} f(x, y, z) = \langle 2x - 3y, 4y - 3x, 2z \rangle$$
$$\vec{\nabla} g(x, y, z) = \langle 2x, 2y, 2z \rangle \qquad \vec{\nabla} h(x, y, z) = \langle 1, 1, 1 \rangle.$$

Setting up the system of equations gives:

$$\begin{aligned} 2x - 3y &= 2x\lambda + \mu \\ 4y - 3x &= 2y\lambda + \mu \\ 2z &= 2z\lambda + \mu \\ x^2 + y^2 + z^2 &= 6 \\ x + y + z &= 0. \end{aligned}$$

Solving this system gives the following candidates on the boundary:

$$\left(\sqrt{6}, -\sqrt{6}, 0\right), \left(-\sqrt{6}, \sqrt{6}, 0\right), \left(\sqrt{3}, \sqrt{3}, -2\sqrt{3}\right),$$
$$\left(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3}\right), \left(0, \sqrt{6}, -\sqrt{6}\right), \left(0, -\sqrt{6}, \sqrt{6}\right).$$

Finally, evaluate f at all candidates:

(x, y, z)	$f(x, y, z)$
$(0, 0, 0)$	0
$\left(\sqrt{6}, -\sqrt{6}, 0\right)$ and $\left(-\sqrt{6}, \sqrt{6}, 0\right)$	12
$\left(\sqrt{3}, \sqrt{3}, -2\sqrt{3}\right)$ and $\left(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3}\right)$	-3
$\left(0, \sqrt{6}, -\sqrt{6}\right)$ and $\left(0, -\sqrt{6}, \sqrt{6}\right)$	6

Thus, the absolute minimum is -3 at both $\left(\sqrt{3}, \sqrt{3}, -2\sqrt{3}\right)$ and $\left(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3}\right)$. The absolute maximum is 12 at both $\left(\sqrt{6}, -\sqrt{6}, 0\right)$ and $\left(-\sqrt{6}, \sqrt{6}, 0\right)$.